

ODD-ORDER TREES WITH EXACTLY THREE VERTICES OF DEGREE TWO ARE EDGE-GRACEFUL

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ABSTRACT. An edge labeling of a graph on n vertices is edge-graceful when the incident edge sums form a complete residue system modulo n . Lee conjectured that every tree of odd order is edge-graceful. A recent preprint proves the conjecture for trees with at most two vertices of degree two. We prove the next case: every odd-order tree with exactly three vertices of degree two is edge-graceful. The main observation is to root the tree at one of those vertices so that the other two are nonadjacent; the problem then reduces to two singleton child blocks. A new two-child common-parent gadget completes the zero-sum block construction. The argument is symbolic for all odd orders at least 25. The remaining 1,061,384 unlabelled trees of odd orders at most 23 are covered by explicit exact certificates with an independent label verifier and a fresh census set-equality check.

1. INTRODUCTION

Let $G = (V, E)$ have p vertices and q edges. An *edge-graceful labeling* is a bijection $f : E \rightarrow \{1, \dots, q\}$ for which

$$f^+(v) = \sum_{e \ni v} f(e) \pmod{p}$$

is a bijection from V to $\mathbb{Z}_p$. Lo introduced this notion in 1985 [4]. For a tree, Lo's necessary congruence forces the order to be odd. Lee conjectured that this is the only obstruction [2].

The zero-sum partition theorem of Kaplan, Lev and Roditty gives the conjecture for trees with at most one vertex of degree two [1]. Meng recently extended this to at most two such vertices using twin zero-sum triples and Langford sequences [7]. The preprint states that the bookkeeping was not pursued for three or more. A different theorem of Lee and Ho concerns trees with three vertices of *even degree*, not three vertices of degree two [3]; it therefore covers only a subclass of the result below.

Theorem 1.1. *Every tree of odd order with exactly three vertices of degree two is edge-graceful.*

The proof uses the packing lemma and nonadjacent gadget rows of Meng [7]. Its new structural step is to use one of the three degree-two vertices as the root. This exposes one parent-capacity case not present in

the earlier catalog, and that case has a four-residue gadget. No proof of Theorem 1.1 or counterexample to it was located in the primary-source searches recorded on 1 September 2026; this is a bounded search statement, not an unconditional priority claim.

2. DEFINITIONS AND REDUCTIONS

All graphs are finite, simple and undirected. Root a tree T of odd order n at r . For $v \neq r$, let $p(v)$ be its parent and let $e^v = vp(v)$. Every edge bijection determines, and is determined by, a bijection $g : V(T) \rightarrow \mathbb{Z}_n$ with $g(r) = 0$ through $g(v) = f(e^v)$. Put

$$B_v = \{g(u) : p(u) = v\}, \quad h(v) = g(v) + \sum_{x \in B_v} x.$$

Then $h(v) = f^+(v)$ modulo n , so f is edge-graceful exactly when h is a bijection.

Lemma 2.1 (degree-two root). *If a tree has exactly three degree-two vertices, one of them can be chosen as a root so that the other two are nonadjacent.*

Proof. The graph induced by the three marked vertices is a forest on three vertices. If it has no edge, choose any marked vertex. If it has one edge, choose an endpoint of that edge. If it has two edges, they form a path, and we choose its middle vertex. The two unchosen vertices are nonadjacent in every case. $\square$

Fix the root r supplied by Lemma 2.1, and call the other two degree-two vertices a, b . The root has two children, while a, b are nonroot internal vertices with unique children a', b' . Every other nonroot internal vertex has at least two children.

Lemma 2.2 (transposition assembly). *Suppose a bijection $g : V(T) \rightarrow \mathbb{Z}_n$, with $g(r) = 0$, satisfies*

$$g(a) = \alpha, \quad g(b) = \beta, \quad g(a') = \beta - \alpha, \quad g(b') = \alpha - \beta,$$

and every child block other than B_a, B_b has sum zero. Then the associated edge labeling is edge-graceful.

Proof. The induced values at a, b are respectively β, α , and every other induced value equals its g -value. Hence h is g followed by a transposition and is a bijection. $\square$

3. ZERO-SUM COMPLETION

Write $K = (n - 1)/2$ and $P_i = \{i, n - i\}$ for $1 \leq i \leq K$. An index triple $i < j < k$ is *admissible* if $i + j = k$ or $i + j + k = n$. In the first case

$$P_i \cup P_j \cup P_k = \{i, j, n - k\} \dot{\cup} \{n - i, n - j, k\},$$

and in the second it is

$$\{i, j, k\} \dot{\cup} \{n - i, n - j, n - k\};$$

each displayed triple has sum zero modulo n . We use the row-aware packing lemma of Meng [7, Lemma 5]: for the residual shapes in the first five rows of Table 1, with the parameter switches specified there when necessary, and also for $[3, K]$, every required number up to the maximum of disjoint admissible index triples exists for every odd $n \geq 27$.

Let $p = p(a)$, $q = p(b)$, $c_v = |B_v|$, and $\delta = \beta - \alpha$. Table 1 prescribes zero-sum cores and, when necessary, a zero-sum host triple in another odd block. The deleted values are assigned to a', b' . The first five rows are the nonadjacent rows of Meng's catalog [7, Table 1]; the final row is new. The mixed row has a mirrored version with a, b interchanged.

TABLE 1. Gadgets for the two nonroot singleton blocks.

Case	Prescribed core(s) and optional host	Deleted	Residual J
$p \neq q$, even/even	$B_p \supset \{1, -1\}; B_q \supset \{3, -3\}$	$\{2, -2\}$	$[4, K]$
$p \neq q$, odd/odd	$B_p \supset \{1, 3, -4\};$ $B_q \supset \{-1, -3, 4\}$	$\{-2, 2\}$	$[5, K]$
$p \neq q$, even/odd	$B_p \supset \{6, -6\}; B_q \supset \{1, 3, -4\};$ host $\{-1, -3, 4\}$	$\{-5, 5\}$	$\{2\} \cup [7, K]$
$p = q$, even ≥ 4	$B_p \supset \{1, -1, 3, -3\}$	$\{2, -2\}$	$[4, K]$
$p = q$, odd	$B_p \supset \{1, 3, -4\};$ host $\{-1, -3, 4\}$	$\{2, -2\}$	$[5, K]$
$p = q$, $c_p = 2$	$B_p = \{1, -1\}$	$\{-2, 2\}$	$[3, K]$

For the primary choices in the six rows, respectively, take

$$(\alpha, \beta) = (1, 3), (1, -1), (6, 1), (1, 3), (1, 3), (1, -1).$$

Thus the two deleted values, in the displayed order up to interchange, are $\delta = \beta - \alpha$ and $-\delta$. This makes the assignment required in Lemma 2.2 explicit. In the parameter-switched large-order cases, the same identity is part of the row-aware statement of the cited packing lemma.

In the new final row set $\alpha = 1$, $\beta = -1$ and $\delta = -2$. The common parent has precisely the two children a, b , its block is zero-sum, and the four residues $1, -1, 2, -2$ are distinct for odd $n \geq 5$. Thus the unused symmetric-pair indices are exactly $[3, K]$.

For every row, the prescribed elements together with the deleted values form a union of complete symmetric pairs. Every core has the parity of its block. A requested host always exists: the sum of the sizes of all nonsingleton blocks is $n - 3$, which is even, so the number of odd such blocks is even. A host row has used one odd block, and any other odd block has size at least

three because the root block has size two and the only singleton blocks are B_a, B_b .

After cores and hosts are placed, suppose $2t$ unfilled odd blocks remain. Their residual capacity is at least $6t$, so $3t \leq |J|$. Choose t disjoint admissible index triples and use their two zero-sum triples in these $2t$ blocks. Fill all remaining even capacities with unused pairs P_i . This assigns every nonzero residue once and makes every nonsingleton block zero-sum.

For $n = 25$, the four needed maximum packings are explicit:

$$\begin{array}{l|l} [3,12] & (3,5,8), (4,7,11), (6,9,10) \\ [4,12] & (4,7,11), (5,8,12), (6,9,10) \\ [5,12] & (5,8,12), (6,9,10) \\ \{2\} \cup [7,12] & (2,9,11), (7,8,10). \end{array}$$

Dropping triples gives every smaller required number. We have therefore proved the following symbolic range.

Proposition 3.1. *Every odd-order tree with exactly three degree-two vertices and order at least 25 is edge-graceful.*

Proof. Choose the root by Lemma 2.1, select the appropriate gadget row, and complete the blocks as above. Lemma 2.2 gives the result. $\square$

4. EXACT FINITE BASE

For odd $5 \leq n \leq 23$, nauty 2.9.3 [5, 6] generated one parent array for each unlabelled tree. Filtering for exactly three degree-two vertices gave the counts in Table 2.

TABLE 2. Certified finite base.

n	5	7	9	11	13	15	17	19	21	23
count	1	3	12	54	267	1,337	6,742	33,906	169,953	849,109

For each of the 1,061,384 trees, an explicit permutation of $\{1, \dots, n-1\}$ was serialized. A fail-closed independent verifier reconstructs the tree, verifies that exactly three vertices have degree two, checks the edge-label bijection, and checks that the vertex residues are exactly $\{0, \dots, n-1\}$. A separate coverage verifier regenerates the nauty census and checks set equality of parent arrays, rejecting duplicate, missing, malformed or extra records. Two order-11 trees outside the gadget constructor were labeled by a separate exhaustive postorder edge search and passed the same verifier. All checks pass.

The complete command is

```
python3 src/verify_all.py
```

It uses only integer arithmetic and reports no solver-indeterminate state. Together with Proposition 3.1, the finite base proves Theorem 1.1.

5. LIMITATIONS AND REPRODUCIBILITY

The theorem is an existence statement, so there is no equality case. It makes no assertion for even order or for four or more degree-two vertices. The infinite construction depends on the packing lemma of the recent preprint [7]; the locally frozen ancillary witnesses and branch checks were rerun, but that preprint is not represented here as peer reviewed. The finite base is a computer-assisted proof whose enumeration trust boundary is nauty's unlabelled-tree generator; label validity is independently checked.

DATA AND CODE AVAILABILITY

The source tree contains the constructor, exact certificates, independent label and census verifiers, structural checker, and one-command orchestrator. The computation used macOS 26.6.1 on arm64, Python 3.9.6, and a workspace-local source build of nauty 2.9.3. No floating-point computation or proof assistant was used.

REFERENCES

1. Gil Kaplan, Arie Lev, and Yehuda Roditty, *On zero-sum partitions and anti-magic trees*, Discrete Mathematics **309** (2009), no. 8, 2010–2014.
2. Sin-Min Lee, *A conjecture on edge-graceful trees*, Scientia, Series A: Mathematical Sciences **3** (1989), 45–47.
3. Sin-Min Lee and Yong-Song Ho, *All trees of odd order with three even vertices are super edge-graceful*, Journal of Combinatorial Mathematics and Combinatorial Computing **62** (2007), 53–64.
4. Sheng-Ping Lo, *On edge-graceful labelings of graphs*, Congressus Numerantium **50** (1985), 231–241.
5. Brendan D. McKay and Adolfo Piperno, *Practical graph isomorphism, II*, Journal of Symbolic Computation **60** (2014), 94–112.
6. ———, *nauty and Traces, version 2.9.3*, Software distribution, 2026, Accessed 1 September 2026.
7. Lingsen Meng, *Trees of odd order with at most two vertices of degree two are edge-graceful*, 2026, Version 1, submitted 24 August 2026.

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